

1 From the electron-phonon interaction to the BCS ground state – overview

The road from deriving the Hamiltonian of the electron–phonon interaction to the BCS Hamiltonian, the physics it contains and finally its ground state is quite long and necessitates many calculations. Hence, this note only gives a concise summary and then refers to the literature for more information. I mostly follow *Solid State Theory 2* by Gerd Czycholl; equation numbers of the form (C 5.7) refer to that book. The Hamiltonian used as a starting point is derived by Czycholl in Sec. 7.1 of his first book and also by Bruus and Flensberg in Ch. 3 of *Many-Body Quantum Theory in Condensed Matter Physics*. If you are simply interested in the BCS ground state, you can directly skip to section 1.4. If you want the details, refer to ??.

1.1 Attractive electron–electron interaction through the electron–phonon mechanism

Intuitive picture/motivation. An electron e_1 moving through the lattice distorts it, i.e. it drags the positive ions towards itself and leaves behind a region of enhanced positive charge. A second electron e_2 then feels an attraction to e_1 through this phonon excitation caused by e_1 . The result is an effective electron–electron interaction that is attractive.

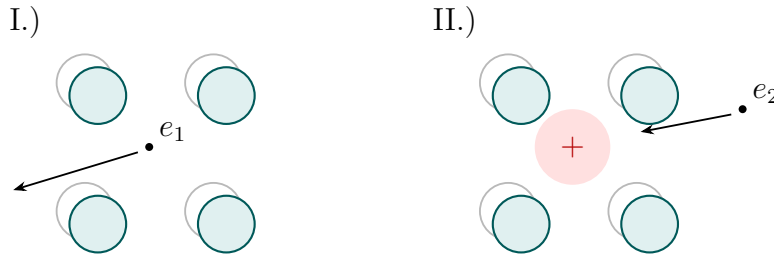


Figure 1: The lattice distortion caused by e_1 (I.) outlives its passage and attracts a second electron e_2 (II.).

Starting point. The starting point for all our derivations is the Fröhlich Hamiltonian, derived in Czycholl Band 1, Sec. 7.1 (7.13)–(7.15) or Bruus and Flensberg, Sec. 3.8, (C 5.7):

$$H = \sum_{\mathbf{k},\sigma} \varepsilon_{\mathbf{k}} c_{\mathbf{k},\sigma}^\dagger c_{\mathbf{k},\sigma} + \sum_{\mathbf{q}} \hbar\omega_{\mathbf{q}} b_{\mathbf{q}}^\dagger b_{\mathbf{q}} + \sum_{\mathbf{k},\mathbf{q},\sigma} \left[M(\mathbf{q}) c_{\mathbf{k}+\mathbf{q},\sigma}^\dagger c_{\mathbf{k},\sigma} b_{\mathbf{q}} + M(-\mathbf{q}) c_{\mathbf{k}-\mathbf{q},\sigma}^\dagger c_{\mathbf{k},\sigma} b_{\mathbf{q}}^\dagger \right], \quad (1)$$

with $\omega_{-\mathbf{q}} = \omega_{\mathbf{q}}$ and $M(-\mathbf{q}) = M(\mathbf{q})$.

Canonical transformation. Use a canonical transformation¹ $H \rightsquigarrow H_T = e^{-iS} H e^{iS}$ to find a purely electronic Hamiltonian that describes this interaction, i.e. to get rid of processes

¹See ?? for more general information on similarity transformations.

that change the number of phonons. At one point one simply restricts the Hamiltonian not to mix states with different phonon configurations.

The canonical transformation yields $H_T \cong H_0 + H_{\text{int}}$ with the interaction (C 5.25)

$$H_{\text{int}} = -\frac{1}{2} \sum_{\sigma, \sigma'} \sum_{\mathbf{k}, \mathbf{k}', \mathbf{q}} V_{\mathbf{k}, \mathbf{k}', \mathbf{q}} c_{\mathbf{k}+\mathbf{q}, \sigma}^\dagger c_{\mathbf{k}'-\mathbf{q}, \sigma'}^\dagger c_{\mathbf{k}', \sigma'} c_{\mathbf{k}, \sigma}, \quad (2)$$

which is attractive if $|\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}-\mathbf{q}}| \leq \hbar\omega_{\mathbf{q}} \leq \hbar\omega_D$ in $V_{\mathbf{k}, \mathbf{k}', \mathbf{q}}$, where ω_D is the Debye frequency, and can be represented as in fig. 2.

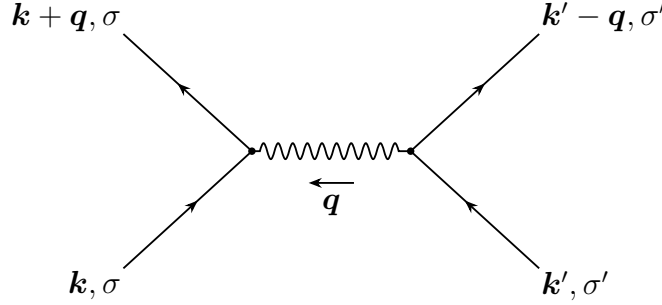


Figure 2: Vertex corresponding to eq. (2): two electrons exchange a phonon of momentum $\mathbf{q}$.

Since we know that the momentum transfer is $\mathbf{q}$, we know that $(\mathbf{k}, \sigma) \rightarrow (\mathbf{k}' - \mathbf{q}, \sigma')$ is not a valid fermion line.

1.2 Cooper pairs

In order to examine how this attractive interaction might change the many-body ground state, one looks at a highly simplified model:

- a system of free, non-interacting electrons, i.e. the Fermi sea (C 5.27)

$$|\phi_0\rangle = \prod_{\varepsilon_{\mathbf{k}} \leq \varepsilon_F, \sigma} c_{\mathbf{k}, \sigma}^\dagger |0\rangle, \quad (3)$$

- the effective interaction is only present for two electrons that are placed right outside of the Fermi sea with opposite momenta and opposite spins (Cooper pair, singlet pairing);
- the interaction is $V = \text{const.}$ if $|\varepsilon_{\mathbf{k}+\mathbf{q}} - \varepsilon_{\mathbf{k}}| \leq \hbar\omega_D$ and zero otherwise.

Here “right outside the Fermi sea” means that the energies are, cf. (C 5.27),

$$E_F \leq \varepsilon_{\mathbf{k}}, \quad \varepsilon_{-\mathbf{k}} \leq E_F + \hbar\omega_D. \quad (4)$$

The interaction thus reads (C 5.28)

$$H_{\text{int}} = -\frac{V}{2} \sum_{\sigma, \sigma'} \sum_{\mathbf{k}, \mathbf{q}} c_{\mathbf{k}+\mathbf{q}, \sigma}^\dagger c_{-\mathbf{k}-\mathbf{q}, \sigma'}^\dagger c_{-\mathbf{k}, \sigma'} c_{\mathbf{k}, \sigma}, \quad \text{with } |\varepsilon_{\mathbf{k}+\mathbf{q}} - \varepsilon_{\mathbf{k}}| \leq \hbar\omega_D. \quad (5)$$

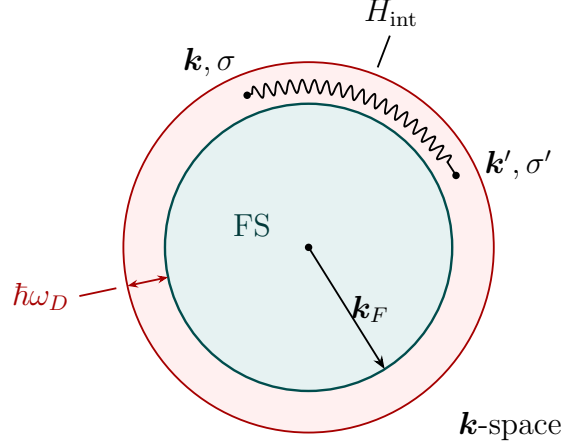


Figure 3: Only states in a shell of width $\hbar\omega_D$ around the Fermi surface take part in the effective interaction.

One solves this model by making the variational ansatz (C 5.26)

$$|\psi_{\sigma,\sigma'}\rangle = \sum_{\mathbf{k}} a_{\sigma,\sigma'}(\mathbf{k}) c_{\mathbf{k},\sigma}^\dagger c_{-\mathbf{k},\sigma'}^\dagger |\phi_0\rangle \quad (6)$$

and minimizing the energy $E = \langle \psi_{\sigma,\sigma'} | H | \psi_{\sigma,\sigma'} \rangle$ with the constraint that $\langle \psi_{\sigma,\sigma'} | \psi_{\sigma,\sigma'} \rangle \stackrel{!}{=} 1$, which gets us an upper bound for the energy of the bound state. One finds that (C 5.35)

$$\Delta E = 2E_F - E_{\text{bs}} \cong -2\hbar\omega_q e^{-2/(V\rho_0)}, \quad (7)$$

which shows that a bound state (Cooper pair) of the two electrons is energetically favoured compared to the free, non-interacting state. This holds for all electrons close to the Fermi edge and therefore implies an instability of the Fermi sea.

1.3 BCS theory

To solve the Hamiltonian and extract the relevant mechanism from it, it is simplified as follows: Only include interactions between electrons with opposite momenta and opposite spins, and neglect the $\mathbf{q}$ -dependence. This yields (C 5.38)

$$H = \sum_{\mathbf{k},\sigma} \varepsilon_{\mathbf{k}} c_{\mathbf{k}\sigma}^\dagger c_{\mathbf{k}\sigma} - V \sum_{\mathbf{k},\mathbf{k}'} c_{\mathbf{k}'\uparrow}^\dagger c_{-\mathbf{k}'\downarrow}^\dagger c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow}, \quad (8)$$

with again $|\varepsilon_{\mathbf{k}} - \varepsilon_{\mathbf{k}'}| \leq \hbar\omega_D$ for the interaction term. Finally, apply mean-field theory to the pair creation/destruction operators, (C 5.41) and (C 5.42):

$$H_{\text{MF}} = \sum_{\mathbf{k}} \begin{pmatrix} c_{\mathbf{k}\uparrow}^\dagger & c_{-\mathbf{k}\downarrow} \end{pmatrix} \begin{pmatrix} \varepsilon_{\mathbf{k}} & -\Delta \\ -\Delta^* & -\varepsilon_{\mathbf{k}} \end{pmatrix} \begin{pmatrix} c_{\mathbf{k}\uparrow} \\ c_{-\mathbf{k}\downarrow}^\dagger \end{pmatrix} \pm \text{const.}, \quad (9)$$

$$\Delta = V \sum_{\mathbf{k}} \langle c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} \rangle \quad (\text{possibly } \varepsilon_{\mathbf{k}} \rightsquigarrow \varepsilon_{\mathbf{k}} - \mu). \quad (10)$$

Bogoliubov transformation. Introduce a unitary transformation that diagonalizes

$$B = \begin{pmatrix} \varepsilon_{\mathbf{k}} & -\Delta \\ -\Delta^* & -\varepsilon_{\mathbf{k}} \end{pmatrix}$$

and thus leads to an H_{MF} that only includes terms of the form $\alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}}$ and $\beta_{\mathbf{k}}^\dagger \beta_{\mathbf{k}}$:

$$H_{\text{MF}} = \sum_{\mathbf{k}} \underbrace{\begin{pmatrix} c_{\mathbf{k}\uparrow}^\dagger & c_{-\mathbf{k}\downarrow} \end{pmatrix} U_{\mathbf{k}}}_{\text{new operators}} \underbrace{U_{\mathbf{k}}^\dagger \begin{pmatrix} \varepsilon_{\mathbf{k}} & -\Delta \\ -\Delta^* & -\varepsilon_{\mathbf{k}} \end{pmatrix} U_{\mathbf{k}}}_{\text{diagonal matrix}} \underbrace{U_{\mathbf{k}}^\dagger \begin{pmatrix} c_{\mathbf{k}\uparrow} \\ c_{-\mathbf{k}\downarrow}^\dagger \end{pmatrix}}_{(\text{new operators})^\dagger}. \quad (11)$$

These new operators will be fermionic, since U is unitary, so that $\langle c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} \rangle \rightsquigarrow f(\alpha_{\mathbf{k}}, \beta_{\mathbf{k}}, \alpha_{\mathbf{k}}^\dagger, \beta_{\mathbf{k}}^\dagger)$ and one can then use $\langle \alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} \rangle = f(E_{\mathbf{k}})$ and $\langle \beta_{\mathbf{k}} \alpha_{\mathbf{k}} \rangle = 0$ etc. The elements of U can be determined by diagonalizing B and requiring that the eigenvectors are normalized to one.

Result. This yields

$$H_{\text{eff}} = \sum_{\mathbf{k}} E_{\mathbf{k}} (\alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} + \beta_{\mathbf{k}}^\dagger \beta_{\mathbf{k}}) + \text{const.}, \quad (\text{C 5.46}) \quad (12)$$

$$E_{\mathbf{k}} = (\varepsilon_{\mathbf{k}}^2 + |\Delta|^2)^{1/2} \geq 0, \quad (\text{C 5.47}) \quad (13)$$

and for Δ one arrives at the BCS self-consistency equation (C 5.50)

$$\Delta = \frac{V\Delta}{2} \sum_{\mathbf{k}} \frac{1}{E_{\mathbf{k}}} \tanh\left(\frac{\beta E_{\mathbf{k}}}{2}\right). \quad (14)$$

Obviously $\Delta = 0$ is a solution. For $\Delta \neq 0$ one obtains (C 5.51)

$$1 = \frac{V}{2} \sum_{\mathbf{k}, |\varepsilon_{\mathbf{k}}| < \hbar\omega_D} (\varepsilon_{\mathbf{k}}^2 + |\Delta|^2)^{-1/2} \tanh\left(\frac{\beta(\varepsilon_{\mathbf{k}}^2 + |\Delta|^2)^{1/2}}{2}\right). \quad (15)$$

Δ as an order parameter. Actually, Δ is an order parameter for the superconducting phase, because of the following:

- For a normal metal $\langle c_{\mathbf{k}\uparrow} c_{-\mathbf{k}\downarrow} \rangle = 0 \ \forall \mathbf{k}$ and thus $\Delta = V \sum_{\mathbf{k}} \langle c_{-\mathbf{k}\downarrow} c_{\mathbf{k}\uparrow} \rangle = 0$.
- But if $\Delta \neq 0$, it means that we are not in a metallic phase and that there are Cooper pairs present.

1.4 The BCS ground state

We have $H = \sum_{\mathbf{k}} E_{\mathbf{k}} (\alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}} + \beta_{\mathbf{k}}^\dagger \beta_{\mathbf{k}})$ with $E_{\mathbf{k}} \geq 0$, cf. eq. (12), where, following (C 5.43),

$$\alpha_{\mathbf{k}} = u_{\mathbf{k}} c_{\mathbf{k}\uparrow} - v_{\mathbf{k}} c_{-\mathbf{k}\downarrow}^\dagger \quad \text{and} \quad \beta_{\mathbf{k}} = u_{\mathbf{k}} c_{-\mathbf{k}\downarrow} + v_{\mathbf{k}} c_{\mathbf{k}\uparrow}^\dagger. \quad (16)$$

Since $\alpha_{\mathbf{k}}^{(\dagger)}$ and $\beta_{\mathbf{k}}^{(\dagger)}$ are fermionic, the eigenvalues of $\alpha_{\mathbf{k}}^\dagger \alpha_{\mathbf{k}}$ (and $\beta_{\mathbf{k}}^\dagger \beta_{\mathbf{k}}$) are 0, 1. Hence the ground state of H does not contain any α or β quasiparticles, i.e.

$$\alpha_{\mathbf{k}}|\text{BCS}\rangle = \beta_{\mathbf{k}}|\text{BCS}\rangle = 0. \quad (17)$$

This is satisfied by

$$|\text{BCS}\rangle = \prod_{\mathbf{k}} (u_{\mathbf{k}} + v_{\mathbf{k}} c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger) |0\rangle = \bigotimes_{\mathbf{k}} (u_{\mathbf{k}} + v_{\mathbf{k}} c_{\mathbf{k}\uparrow}^\dagger c_{-\mathbf{k}\downarrow}^\dagger) |0\rangle_{\mathbf{k}}. \quad (18)$$

1.5 References

- G. Czycholl, *Theoretische Festkörperphysik* Band 1, Sec. 7.1 (Fröhlich Hamiltonian) and Band 2, Ch. 5 (Cooper pairs, BCS theory).
- H. Bruus & K. Flensberg, *Many-Body Quantum Theory in Condensed Matter Physics*, Ch. 3.
- J. Bardeen, L.N. Cooper & J.R. Schrieffer, Phys. Rev. **108**, 1175 (1957).